

**M10 / ASSESSMENT**

# M10 Module Revision Test

**Standalone Zero-to-Job-Ready Data Engineer 2026**

Study, practise, check and explain.

**A complete learning route**

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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## Read this first

### Closed book - no forced timer

Save the full attempt before opening protected answers. The test samples every canonical lesson; G04 remains the stronger pipeline boundary assessment.

### 1. DE10-01 - Why containers exist

Write a before/after environment story for `worker.py`: list what must be installed/configured without Docker, then list what moves into the image and what deliberately remains runtime configuration. Explain one problem containers solve and one problem they do not solve.

### 2. DE10-02 - Images and containers

Draw an image-to-container one-to-many diagram. For five changes - Python base version, application source, input URL, host port and API token - classify whether each normally belongs to image build, runtime configuration or secret management.

### 3. DE10-03 - Writing a Dockerfile

Explain every instruction in the supplied Dockerfile in one sentence. Then modify a working copy so `worker.py` is copied to `/opt/m10/worker.py`; update `WORKDIR/CMD` consistently and predict what breaks if only one path is changed.

### 4. DE10-04 - Ports

Change the host port to 8095 without changing the container port. State the host URL and the worker's internal URL. Explain why the worker should not start calling `localhost:8095`.

### 5. DE10-05 - Volumes and persistence

Classify five paths - immutable input CSV, generated report, application source during development, database data directory, secret file - as image content, bind mount, named volume or secret mechanism. Defend each choice.

### 6. DE10-06 - Networks and service communication

Explain three addresses: `localhost:8090` from the host, `source:8000` from worker, and `localhost:8000` from inside source. State which namespace each `localhost` refers to.

### 7. DE10-07 - Compose, configuration and secrets

From a blank page, reconstruct a two-service Compose design: source + worker, internal service-name URL, one host port, read-only input and named output volume. Then compare to the supplied file and record only differences that matter.

### 8. DE10-08 - Why orchestration exists

Draw extract -> validate -> publish for one date. Mark which step owns data correctness and which orchestration concerns are added around it: dependency, retries, run metadata and historical rerun.

## 9. DE10-09 - Airflow architecture and UI

Write a troubleshooting decision tree for: Dag absent from UI, Dag present but no scheduled run, run exists but one task fails. Name the first evidence you would inspect in each case.

## 10. DE10-10 - DAGs, tasks and dependencies

Redesign the graph so after validation it builds `finance_report` and `ops_report` in parallel, then one handoff waits for both. Draw it before writing syntax and state why each edge exists.

## 11. DE10-11 - TaskFlow, operators and task communication

For five candidate values - partition date, 2 GB dataframe, output table name, validation row count, API password - decide XCom/small return, durable storage reference, Connection/secret, or something else. Defend each boundary.

## 12. DE10-12 - Scheduling and data intervals

Given three runs representing Sep 1, Sep 2 and Sep 3 but physically executed on Sep 10, write the exact input path each must use. Then explain why wall-clock code would make all three collide.

## 13. DE10-13 - Retries, timeouts and failure behavior

Classify timeout, HTTP 503, missing required column, bad password, temporary network reset and negative quantity as retry/repair/fail categories. Explain where idempotency changes your decision.

## 14. DE10-14 - Parameters and task communication

Design Params for `min_rows`, `target_region` and `dry_run`. Give type/default/validation rules. Then list which facts should instead be task results: observed rows, output path, actual total.

## 15. DE10-15 - Connections, variables and secrets

Classify ten values: warehouse password, warehouse connection ID, alert threshold, partition date, output row count, Slack token, static table name, run-specific `dry_run`, source system URL, 2 GB result set. State where each belongs.

## 16. DE10-16 - Sensors and waiting

Choose sensible poke interval and timeout for three cases: file expected within 5 minutes, partner file within 2 hours, daily data warehouse table expected by 08:00. Defend resource vs detection-latency tradeoffs.

## 17. DE10-17 - Logs and debugging

Take a simulated failed partition and write a six-field incident note: Dag/run, task, partition, attempt, observed error, next evidence to inspect. Then list three fields that should never be logged raw in a real system.

## 18. DE10-18 - Catchup, backfills and reprocessing

Design a backfill for Sep 1-7 where Sep 4 is known bad. State input/output partition naming, failure policy, whether later dates continue, and how you will reconcile after repair.

## 19. DE10-19 - Integration and production-style orchestration

Write a one-page operational handoff: clean setup, config/secrets ownership, Dag partition rule, retry/timeout policy, sensor/wait policy, data validation, backfill steps, evidence locations and rollback/recovery boundary. Another learner should be able to explain the design without your verbal help.